Exact rank-one encoding with a partial-trace decoder for the Li–Jiang two-qubit channel

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Abstract

For the two-qubit noise family of Li and Jiang, we determine the exact maximum entanglement fidelity over all rank-one completely positive trace-nonincreasing encoders when the decoder is the partial trace. For every $0 < p < 1$, an optimizer has columns $x|00\rangle + y|11\rangle$ and $|10\rangle$ with real $x \geq 0$, $y \leq 0$. The optimal fidelity is an explicit algebraic function specified by a unique quartic root. The proof reduces the full complex encoder problem to a trace-norm maximization and establishes alignment by a convex determinant identity. A multistart numerical search with arbitrary CPTP decoders finds no improvement on a 100-point noise grid; joint rank-one optimality remains open.

1 The question and channel

Li and Jiang construct a noise family for which higher-rank encoding is necessary to attain the unrestricted optimum [1]. Their polar rank-one encoder attains the optimal quadratic gap coefficient as $p \to 0^+$, while finite-noise corrections remain undetermined after Eq. (S64). We determine whether the two-column family considered below attains the full rank-one optimum with the partial-trace decoder. We prove that it does, including complex and trace-nonincreasing encoders, and give its exact optimum.

We specialize Eqs. (18)–(23) of Ref. [1] to $L = S = \mathbb{C}^2$, $C = L \otimes S$, and $\tau = I_2/2$. The code basis is $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. Put

$$\begin{gathered} \psi = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad \Pi = |\psi\rangle\langle\psi|, \qquad D_j = I_2 \otimes \langle j|, \\ \lambda_1 = 1 - \frac{p}{2}, \quad \lambda_2 = \frac{p(3-2p)}{2(3+p)}, \quad \lambda_3 = \frac{p^2}{2(3+p)}, \quad \lambda_4 = \frac{3+p}{4}, \quad \lambda_5 = \frac{1-p}{4}, \\ a = \sqrt{\frac{3+p}{6}}, \quad b = \sqrt{\frac{1-p}{2}}, \quad Q = a(I_4 - \Pi) + b\Pi. \end{gathered} \tag{1}$$

For $i, j \in \{0, 1\}$ define

$$E_{ij} = \frac{|ij\rangle\langle\psi| - \sqrt{\lambda_5} D_j^\dagger D_i Q}{\sqrt{\lambda_4}}.$$

The literal noise action is

$$\mathcal{N}_p(X) = \lambda_3 \operatorname{tr}(X) I_4 + (\lambda_1 - \lambda_3) \operatorname{tr}_S(QXQ) \otimes I_2 + \lambda_2 \sum_{i,j} E_{ij} X E_{ij}^\dagger. \tag{2}$$

For a logical map with Kraus operators L_α, our convention is

$$F_{\mathrm{e}}(\tau,\Phi)=\frac{1}{4}\sum_{\alpha}|\operatorname{tr}L_\alpha|^2. \tag{3}$$

This is the unrenormalized entanglement fidelity, including for maps that are trace-nonincreasing. A rank-one encoder is $\mathcal{C}_V(X)=VXV^\dagger$, $V\in\mathbb{C}^{4\times 2}$, $V^\dagger V\preceq I_2$. Write

$$F_*(p)=\max_{V^\dagger V\preceq I_2} F_{\mathrm{e}}(\tau,\operatorname{tr}_S\circ\mathcal{N}_p\circ\mathcal{C}_V).$$

2 Exact optimum

Theorem 1. *For every $0<p<1$, the maximum $F_*(p)$ is attained by*

$$V(x,y)=\begin{pmatrix}x&0\\0&0\\0&1\\y&0\end{pmatrix},\qquad x^2+y^2=1,\quad x>0,\ y<0. \tag{4}$$

Define

$$k=\frac{3-2p}{3+p},\qquad h=\frac{1+p}{2},\qquad w=\sqrt{\frac{(3+p)(1-p)}{12}},$$
$$A=k(h+w),\qquad B=k(h-w). \tag{5}$$

Let $\eta>A$ be the unique root in that interval of

$$\eta^4-2(A+B)\eta^3+4AB\eta^2-(AB)^2=0. \tag{6}$$

Then

$$\boxed{F_*(p)=\frac{1}{4}\left(1+\eta+\frac{A^2}{\eta-A}+\frac{B^2}{\eta-B}\right)},\qquad x_*=\frac{A}{\eta-A},\quad y_*=-\frac{B}{\eta-B}. \tag{7}$$

Equivalently, $\eta=k\mu$, where $\mu>h+w$ is the unique root there of

$$9\mu^4-18(1+p)\mu^3+12p(p+2)\mu^2-p^2(p+2)^2=0. \tag{8}$$

2.1 The quadratic form and contraction reduction

A Kraus family for (2) consists of

$$\sqrt{\lambda_3}|u\rangle\langle v|,\qquad \sqrt{\lambda_1-\lambda_3}(I_2\otimes|s\rangle\langle t|)Q,\qquad \sqrt{\lambda_2}E_{ij},$$

with respectively 16, 4, and 4 operators. Put

$$C_i=QD_i^\dagger,\quad S_i=|\psi\rangle\langle i|,\quad H_i=\frac{S_i-\sqrt{\lambda_5}C_i}{\sqrt{\lambda_4}},\quad \beta=\frac{\lambda_1-\lambda_3}{2},\quad \delta=\frac{\lambda_2}{2}.$$

The identity

$$D_\ell E_{ij}=\frac{\delta_{\ell j}}{\sqrt{\lambda_4}}\left(|i\rangle\langle\psi|-\sqrt{\lambda_5}D_iQ\right)$$

and the Kraus trace formula give, for every complex V,

$$F_p(V) = \frac{\lambda_3}{2}\|V\|_{\mathrm{HS}}^2 + \beta \sum_i |\langle C_i, V\rangle_{\mathrm{HS}}|^2 + \delta \sum_i |\langle H_i, V\rangle_{\mathrm{HS}}|^2. \tag{9}$$

Here $\langle X, Y\rangle_{\mathrm{HS}} = \mathrm{tr}(X^\dagger Y)$. In particular, this is a positive semidefinite quadratic form on the full eight-dimensional complex encoder space. Directly,

$$\langle C_i, C_j\rangle_{\mathrm{HS}} = \langle H_i, H_j\rangle_{\mathrm{HS}} = \delta_{ij}, \qquad \langle C_i, H_j\rangle_{\mathrm{HS}} = 0. \tag{10}$$

For example, $Q^2 = (3+p)I_4/6 - (2p/3)\Pi$ yields the first identity, and $\langle C_i, S_j\rangle_{\mathrm{HS}} = \sqrt{\lambda_5}\delta_{ij}$ yields the others.

For completeness, the contraction reduction of Lemma S1 in Ref. [1] has a short proof here. Extend the polar factor of a contraction to an isometry U, so $V = UT$, $0 \preceq T \preceq I_2$. The two matrices

$$V_\pm = U\big(T \pm i\sqrt{I_2 - T^2}\big)$$

are isometries with $V = (V_+ + V_-)/2$. For a positive quadratic form q,

$$\frac{q(V_+) + q(V_-)}{2} - q(V) = \frac{1}{4}q(V_+ - V_-) \geq 0.$$

Thus at least one isometry has fidelity no smaller than V. It suffices to maximize (9) over $V^\dagger V = I_2$, where the first term is λ_3.

2.2 Trace norm and alignment

For spinors $r, s \in \mathbb{C}^2$ define

$$C(r) = \sum_i r_i C_i, \qquad H(s) = \sum_i s_i H_i, \qquad M(r,s) = \sqrt{\beta}C(r) + \sqrt{\delta}H(s).$$

Cauchy–Schwarz duality followed by the polar decomposition gives

$$\begin{aligned} F_*(p) - \lambda_3 &= \max_{\|r\|^2+\|s\|^2=1} \max_{V^\dagger V = I_2} |\,\mathrm{tr}(M(r,s)^\dagger V)|^2 \\ &= \max_{\|r\|^2+\|s\|^2=1} \|M(r,s)\|_1^2. \end{aligned} \tag{11}$$

Here $\|M\|_1$ is the sum of its singular values. The polar factor attains the inner maximum, extending to an isometry if needed.

For $U \in U(2)$, set $W_U = U \otimes \overline{U}$. Since $W_U\psi = \psi$ and $W_U Q = Q W_U$,

$$W_U C(r) U^\dagger = C(\overline{U}r), \qquad W_U H(s) U^\dagger = H(\overline{U}s).$$

These identities preserve trace norm. Hence, for fixed norms R, S, we may write $r = R(1,0)^T$, $s = S(c,d)^T$, with $d \geq 0$, $|c|^2 + d^2 = 1$. The case $R = 0$ is covered by the same choice using the full unitary freedom. Write $c = X + iY$ and put

$$t = \frac{b}{a} = \sqrt{\frac{3(1-p)}{3+p}} \in (0,1), \quad u = \sqrt{\beta}R, \quad v = \sqrt{\delta}S.$$

Thus $M = uC_0 + v[(X + iY)H_0 + dH_1]$. Direct multiplication, using $d^2 = 1 - X^2 - Y^2$, yields

$$\begin{aligned} \operatorname{tr}(M^\dagger M) &= u^2 + v^2, \\ \det(M^\dagger M) &= D_0 + D_1 X + D_x X^2 + D_y Y^2, \\ D_0 &= \frac{18(1+t^2)u^4 + 2t^2(t^2+9)v^4 + 3(t^4 - 2t^2 + 9)u^2v^2}{9(t^2+3)^2}, \\ D_1 &= \frac{4\sqrt{3}\, tuv[3(1-t^2)u^2 - (9-t^2)v^2]}{9(t^2+3)^2}, \\ D_x &= \frac{u^2v^2(1-t^2)(9-t^2)}{3(t^2+3)^2} \geq 0, \qquad D_y = \frac{u^2v^2}{3} \geq 0. \end{aligned} \tag{12}$$

This proves that the determinant is convex on the disk $X^2 + Y^2 \leq 1$. Its maximum occurs on the boundary: express any interior point as a convex combination of boundary points and use convexity. Since the trace is fixed and

$$\|M\|_1^2 = \operatorname{tr}(M^\dagger M) + 2\sqrt{\det(M^\dagger M)},$$

a trace-norm maximizer also occurs on the boundary. Therefore $d = 0$ and r, s may be chosen collinear. Their relative phase need not be fixed in this step.

For collinear spinors the first column of M lies in $\operatorname{span}\{|00\rangle, |11\rangle\}$ and the second lies in $\operatorname{span}\{|10\rangle\}$. Its polar isometry has the same form, with arbitrary isometric extensions chosen in these spaces when a column vanishes. After an irrelevant common phase it is $V(x, y)$ with complex x, y satisfying $|x|^2 + |y|^2 = 1$. Choose a maximizing collinear pair (r, s) and such a polar isometry V. The inner product in (11) then has squared modulus $F_*(p) - \lambda_3$. Duality gives $F_p(V) - \lambda_3$ at least this value, so this complex two-column family attains the full maximum.

2.3 Phases and the circle maximum

Substitution in (9) and use of $|x|^2 + |y|^2 = 1$ give

$$F_p(V(x, y)) = \frac{1}{4}\left[1 + A(|x|^2 + 2\Re x) + B(|y|^2 - 2\Re y)\right]. \tag{13}$$

The coefficients obey $A > B > 0$, since $k > 0$, $w > 0$, and $h^2 - w^2 = p(p+2)/3 > 0$. Thus replacing x by $|x|$ and y by $-|y|$ cannot decrease fidelity. Set $U = x \geq 0$, $W = -y \geq 0$, $U^2 + W^2 = 1$. For every $\eta > A$,

$$\begin{aligned} &\eta + \frac{A^2}{\eta - A} + \frac{B^2}{\eta - B} - [A(U^2 + 2U) + B(W^2 + 2W)] \\ &\qquad = (\eta - A)\left(U - \frac{A}{\eta - A}\right)^2 + (\eta - B)\left(W - \frac{B}{\eta - B}\right)^2. \end{aligned} \tag{14}$$

This is a matching global upper bound when

$$\frac{A^2}{(\eta - A)^2} + \frac{B^2}{(\eta - B)^2} = 1. \tag{15}$$

The left side decreases strictly from infinity to zero on (A, ∞), so there is exactly one solution there. Clearing its positive denominators gives (6); substitution gives (7). Using $A + B = k(1 + p)$ and $AB = k^2 p(p + 2)/3$ gives (8). This completes the proof of Theorem 1.

3 Checks, decoder optimization, and scope

At $p = 9/13$, $A = 105/208$, $B = 49/208$, and

$$x_* \approx 0.9553880907, \qquad y_* \approx -0.2953533412, \qquad F_*(9/13) \approx 0.6462632754.$$

On the real circle, Eq. (13) is equivalently

$$F_{9/13}(V(x,y)) = \frac{132x^2 + 210x + 76y^2 - 98y + 181}{832}.$$

The polar choice $(3/\sqrt{10}, -1/\sqrt{10})$ has fidelity $0.6461704491\ldots$, and $(24/25, -7/25)$ has fidelity $0.6462134615\ldots$. Both are strictly below the new maximum. The endpoint checks are

$$F_*(p) = 1 - \frac{p}{2} - \frac{p^2}{36} + O(p^3), \qquad \lim_{p\to 1^-} F_*(p) = \frac{5 + 2\sqrt{2}}{16}. \tag{16}$$

The first follows from $A = 1 - 2p/3 - p^2/9 + O(p^3)$, $B = 2p/3 + O(p^2)$, $W_* = p/3 + O(p^2)$, and $U_* = 1 - p^2/18 + O(p^3)$. It agrees with the quadratic gap in Ref. [1].

We also optimized an arbitrary CPTP decoder numerically. This decoder class suffices for the joint optimum: completing any CPTNI decoder by adding $X \mapsto \mathrm{tr}[(I_4 - \mathcal{D}^\dagger(I_2))X]|0\rangle\langle 0|$ makes it trace preserving and cannot decrease the unrenormalized fidelity (Ref. [1], Sec. S3.D). For fixed V, put $L_\alpha = K_\alpha V$ and $g_\alpha = \mathrm{vec}_{\mathrm{row}}(L_\alpha^T)$. If $J_\mathcal{D} = \sum_\ell \mathrm{vec}_{\mathrm{row}}(D_\ell)\,\mathrm{vec}_{\mathrm{row}}(D_\ell)^\dagger$, then

$$R_V = \frac{1}{4}\sum_\alpha \overline{g_\alpha} g_\alpha^T, \qquad \max_{J_\mathcal{D}\succeq 0,\ \mathrm{tr}_{\mathrm{out}} J_\mathcal{D} = I_4} \mathrm{tr}(J_\mathcal{D} R_V) = \min_{I_2\otimes Y \succeq R_V} \mathrm{tr}\, Y. \tag{17}$$

Here Y is Hermitian. We alternated this decoder SDP with polar Reimpell–Werner encoder updates, as in Algorithm 2 of Ref. [1]. At each of $p = k/100$ ($1 \le k \le 99$) and $p = 9/13$, we used eight starts: the encoder in (7) and seven independent complex Gaussian polar factors. Decoder SDPs used CLARABEL; the maximum observed primal/adjusted-dual gap was below 3.9×10^{-9}. These are floating-point bounds, and the saved encoder isometry and decoder trace-preservation residuals were below 3×10^{-15}. All 800 runs stopped below a 2×10^{-10} outer fidelity increment. No decoder improvement over $F_*(p)$ was found within 10^{-8}. Representative exact-formula values are

p	0.10	0.25	0.50	0.75	0.90
$F_*(p)$	0.949740515	0.873544787	0.745246955	0.616635242	0.539960567

These finite computations support equality of the joint rank-one optimum and $F_*(p)$; they do not prove it. A uniform bound with a variable decoder remains open.

Li and Jiang prove that the unrestricted optimum is $\lambda_1 = 1 - p/2$ and every attaining pair has encoder Choi rank at least two for $0 < p < 1$ [1]. Theorem 1 gives an exact fixed-decoder rank-one benchmark for their example.

Accompanying scripts check Kraus completeness, orthogonality, and the quadratic, determinant, family, and quartic identities in exact arithmetic with $p = 3(1 - t^2)/(3 + t^2)$. Reproduction details and numerical witnesses with feasibility residuals accompany the note. The proof is independent of the numerical search.

References

[1] B. Li and L. Jiang, *High-Rank Encoding Can Improve Approximate Quantum Error Correction*, arXiv:2609.00778v1 (2026), `https://arxiv.org/abs/2609.00778v1`.